

STATISTICAL HYPOTHESIS STATISTICAL TESTING

Module 2

HYPOTHESIS

- A **hypothesis** is an assumption about population parameter to be tested based on sample information
- Hypothesis are general statements about the probability distribution based on assumptions that may or may not be true
- If θ be the parameter of the population and $\hat{\theta}$ is the estimate of θ in the random sample drawn from population, then the difference between θ and $\hat{\theta}$ is called sampling fluctuations and should be small or insignificant.
- If the difference between θ and $\hat{\theta}$ is high then the difference is significant.
- Significant difference arises due to the fact that the either sampling procedure is not purely random or sample is not from the population.
- If the sample statistics differs from the hypothesis made about population parameter, a decision is made to test whether the difference is significant or not.
- If difference is significant then hypothesis is rejected, if not it is accepted.
- This process is called **Hypothesis Testing**.

PROCEDURE OF HYPOTHESIS TESTING

1. State H_0 and H_1
2. Select α
3. Choose test
4. Calculate test statistic
5. Make decision

STATE H_0 AND H_1

First step is to state two hypothesis H_0 and H_1 so that if one true then other is false and vice versa.

Null Hypothesis H_0 asserts that there is no true difference between sample statistics and population parameter under consideration and differences found are due to sample fluctuation and are insignificant.

That is, there is no difference between assumed and actual values

Alternate Hypothesis H_1 is reverse of H_0 , that is, there is significant difference between assumed and actual values.

Accepting H_0 indicated that there is not enough evidence to show H_1 is true.

There are three type of possibilities for stating hypothesis

$$H_0: \mu \geq \mu_0$$

$$H_1: \mu < \mu_0$$

$$H_0: \mu \leq \mu_0$$

$$H_1: \mu > \mu_0$$

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

Where μ and μ_0 are population and sample parameters

The hypothesis testing attempts to find enough evidence for establishing alternative hypothesis H_1

SELECT SIGNIFICANT LEVEL α

Significant level α defines the confidence with which null hypothesis H_0 has been accepted or rejected after testing

Generally α is taken either 5% or 1%

If $\alpha = 0.05$ then there are about 5 chances out of 100 that null hypothesis H_0 is rejected when it should have been accepted. Researchers are 95% confident that results are right and test result is said to be “**significant**”.

If $\alpha = 0.01$ then there are about 1 chance out of 100 that null hypothesis H_0 is rejected when it should have been accepted. Researchers are 99% confident that results are right and test result is said to be “**highly significant**”.

CHOOSE TEST

The third step is to choose suitable test statistics and its distribution based on:

- Sample size (n)
- Known/unknown population variance
- Type of data
- Number of samples

Common tests are:

- **Z-test** → Large sample ($n \geq 30$), σ known
- **t-test** → Small sample, σ unknown
- **Chi-square test** → Categorical data
- **F-test / ANOVA** → Comparing more than two means

CALCULATE TEST STATISTIC

Next step is to perform calculations on random sample necessary for the chosen test and check whether the result lies in critical region or in acceptance region.

Values that lead to rejection of null hypothesis are ***critical region*** of the distribution.

The rest values are acceptance region.

MAKE DECISION

Decision is made to either accept or reject null hypothesis

HYPOTHESIS ERRORS

When hypothesis is tested, there are four possible results

1. hypothesis is true but test rejects it
2. hypothesis is false but test accepts it
3. hypothesis is true and test accepts it
4. hypothesis is false and test rejects it

Decisions

Accept H_0

Reject H_0

Conditions

H_0 : True

H_1 : False

Correct Decision	Type II Error
Type I Error	Correct Decision

The first two possibilities lead to errors.

- If hypothesis is rejected when it should be accepted (False Positive), it is said that **type I error** has occurred
- If hypothesis is accepted when it should be rejected (False Negative), it is said that **type II error** has occurred

The significance level (α) is the probability of committing this error.

- $P(\text{Type I Error}) = \alpha$

Level of confidence = $1 - \alpha$

- $P(\text{Type II Error}) = \beta$

Power of a Test = $1 - \beta$

Power represents the probability of correctly rejecting a false H_0 .

Higher power $\rightarrow$ Better test.

Example:

H_0 : Patient is healthy

H_1 : Patient is diseased

Type I Error $\rightarrow$ Healthy person diagnosed as diseased

Type II Error $\rightarrow$ Diseased person diagnosed as healthy

ONE- TAILED TEST

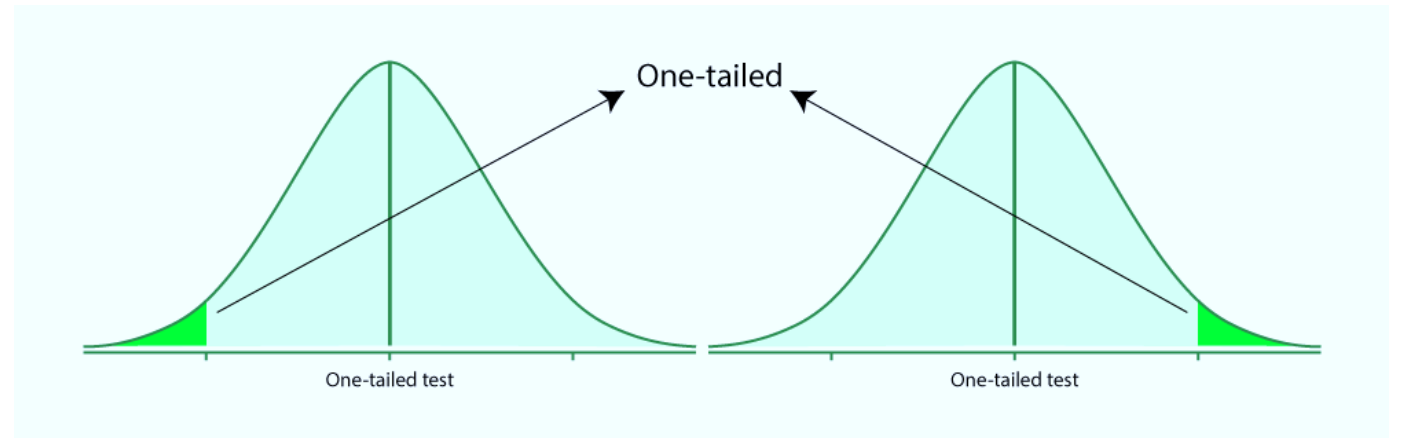
A one-tailed test is based on a unidirectional hypothesis where the area of rejection is on only one side of the sampling distribution. It determines whether a particular population parameter is larger or smaller than the predefined parameter. It uses one single critical value to test the data.

Null Hypothesis (H_0): where θ represents a parameter (e.g., population mean) and θ_0 is a specific value.

Alternative Hypothesis (H_1):

For a right-tailed test: $H_1: \theta > \theta_0$

For a left-tailed test: $H_1: \theta < \theta_0$



TWO TAILED TEST

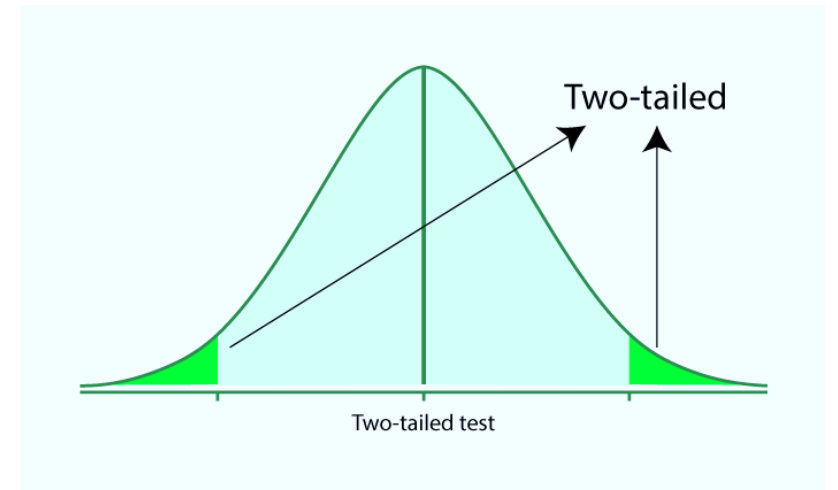
Two-tailed test is where the hypothesis about the population mean is rejected for value of falling into either tail of the sampling distribution.

Null Hypothesis (H_0): where θ represents a parameter (e.g., population mean) and θ_0 is a specific value.

Alternative Hypothesis (H_1): Alternative Hypothesis (H_1): $\theta \neq \theta_0$ (not equal to)

Test Statistic: Compute the test statistic as appropriate for the distribution (Z-score for normal distribution).

Decision Rule: If the test statistic falls in either tail of the distribution's critical region, reject the null hypothesis in favour of the alternative hypothesis.



Z TEST

A **Z-test** is a statistical hypothesis test used to determine whether there is a **significant difference between a sample statistic and a population parameter** when:

- The **population variance (σ^2) is known**, or
- The **sample size is large ($n \geq 30$)** so the **sampling distribution is approximately normal** according to the **Central Limit Theorem**.

Decision Rule

- If **$|Z \text{ calculated}| > Z \text{ critical} \rightarrow \text{Reject } H_0$**
- If **$|Z \text{ calculated}| \leq Z \text{ critical} \rightarrow \text{Fail to Reject } H_0$**

Critical Z-values			
Level of Significance α	0.10	0.05	0.01
Z-value for one tailed test	+1.28 Or -1.28	+1.645 Or - 1.645	+2.33 Or - 2.33
Z-value for two tailed test	± 1.645	± 1.96	± 2.58

It is used to test hypotheses about:

Hypothesis Test	Z-Test Formula	Parameters
Population Mean (One-sample Z-test)	$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$\bar{X}$ =sample mean, μ = population mean, σ = population standard deviation, n = sample size
Difference Between Two Means	$z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} - \frac{\sigma_2^2}{n_2}}}$	$\bar{X}_1, \bar{X}_2$ = sample means, $\sigma_1 \sigma_2$ = population SDs, $n_1 n_2$ = sample size
Population Proportion	$z = \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}}$	p = sample proportion, P = population proportion, n = sample size
Difference Between Two Proportions	$z = \frac{p_1 - p_2}{\sqrt{P(1-P)(\frac{1}{n_1} + \frac{1}{n_2})}}$	$p_1 p_2$ = sample proportions, P = population proportionSDs, $n_1 n_2$ = sample size

Example (population Mean): A manufacturer of cheaper type of the light bulb claims that his product is better than the high priced competitive light bulb. The average life of other light bulb is known to be 5000 hours. In a test to examine the manufacturers claim, 100 of his bulbs are left on until they burn out, the average length of life in the sample is 5100 hours. With a significance level of $\alpha = 0.05$ is there enough evidence to support the manufacturer's claim ? Assume that $\sigma = 500$ hours.

Hypothesis: $H_0: \mu = 5000, H_1: \mu > 5000$

$\bar{X} = 5100$ (Sample mean), $\mu = 5000$ (population mean), $\sigma = 500$ (population standard deviation), $n = 100$ (sample size)

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{5100 - 5000}{500 / \sqrt{100}} = \frac{100}{50} = 2.0$$

At $\alpha = 0.05$ critical $z = 1.645$

| Z calculated | > Z critical i.e. $2.0 > 1.645 \rightarrow$ Reject H_0 and accept H_1

Therefore, it is concluded that manufacturers claim is justified

Example (difference between two means): The two types of batteries are tested for their length of life and following results are obtained. Can we conclude that two types of batteries are having the same mean life?

Battry	Sample size	Mean (hours)	Standard deviation (hours)
ABC	100	1000	10
XYZ	120	1050	11

Hypothesis:

$H_0: \mu_1 = \mu_2$ The two types of batteries have the same mean life.

$H_1: \mu_1 \neq \mu_2$ The two types of batteries have different mean lives.

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{1000 - 1050}{\sqrt{\frac{(10)^2}{100} + \frac{(11)^2}{120}}} = -\frac{50}{\sqrt{2.0083}} = -35.28$$

At $\alpha = 0.05$, $|Z \text{ calculated}| > Z \text{ critical}$ i.e. $35.28 > 1.965 \rightarrow \text{Reject } H_0 \text{ and accept } H_1$

Hence, the mean life differs significantly

Example (test of number of success): In 600 throws of six-faced die, odd points appear 360 times. Would you say dice is fair?

$$n = 600$$

$$P = 0.5$$

$$\text{Mean of sampling distribution} = \mu = np = 600 \times 0.5 = 300$$

$$\text{Standard Deviation } \sigma = \sqrt{np(1 - p)} = \sqrt{600 \times 0.5 \times 0.5} = 12.25$$

Hypothesis:

$H_0: \mu_1 = \mu_2$ The die is fair, so the probability of getting an odd number is 0.5.

$H_1: \mu_1 \neq \mu_2$ The die is **not fair**, so the probability of getting an odd number is different from 0.5.

$$Z = \frac{\bar{X} - \mu}{\sigma} = \frac{360 - 300}{12.25} = 4.9$$

At $\alpha = 0.05$, $|Z \text{ calculated}| > Z \text{ critical}$ i.e. $4.9 > 1.965 \rightarrow \text{Reject } H_0 \text{ and accept } H_1$

Conclusion: The die is not fair

Example (population proportion): in sample of 800 parts manufactured by a company, number of defected part are found to be 60. However company claims that only 7% of their product are defective. Apply appropriate test to verify company's claim.

Hypothesis:

$H_0: p = P = 0.07$

The proportion of defective parts is **7%**.

The company's claim is **true**.

$H_1: p > P = 0.07$

The proportion of defective parts is **greater than 7%**.

The company's claim is **not justified**.

$P = 0.07$

$$p = \frac{60}{800} = 0.075$$

$$Z = \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}} = \frac{0.075 - 0.07}{\sqrt{\frac{0.07 \times 0.93}{800}}} = \frac{0.005}{0.009} = 0.556$$

At $\alpha = 0.05$, $|Z \text{ calculated}| < Z \text{ critical}$ i.e. $0.556 < 1.645 \rightarrow \text{Accept } H_0$

Hence, company's claim is justified

Example (difference between two proportions): on a random sample of 100 persons taken from village A, 60 are found to be consuming tea. In another sample of 200 persons taken from a village B, 100 persons found to be consuming tea. Do the data reveal significant difference between two villages as far as the habit of consuming tea is concerned?

Hypothesis:

$$H_0: p_1 = p_2$$

There is **no significant difference** in the proportion of tea consumers in villages A and B.

$$H_1: p_1 \neq p_2$$

There is **a significant difference** in the proportion of tea consumers in villages A and B.

$$p_1 = \frac{x_1}{n_1} = \frac{60}{100} = 0.6, \quad p_2 = \frac{x_2}{n_2} = \frac{100}{200} = 0.5$$

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{60 + 100}{100 + 200} = 0.53$$

$$Z = \frac{p_1 - p_2}{\sqrt{P(1-P)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.6 - 0.5}{\sqrt{(0.53)(0.47)\left(\frac{1}{100} + \frac{1}{200}\right)}} = \frac{0.1}{\sqrt{0.0037}} = \frac{0.1}{0.0608} = 1.64$$

At $\alpha = 0.05$, $|Z \text{ calculated}| < Z \text{ critical i.e. } 1.64 < 1.96 \rightarrow \text{Accept } H_0$

There is no significant difference between two villages as far as the habit of consuming tea is concerned

T TEST

A **t-test** is a statistical hypothesis test used to determine whether there is a **significant difference between sample means** when the **population standard deviation is unknown** and the **sample size is small ($n < 30$)**.

The test is based on the **Student's t-distribution**, which is similar to the **Standard Normal Distribution** but has **heavier tails** to account for additional uncertainty from estimating the population standard deviation.

Use a t-test when:

- Population **standard deviation (σ) is unknown**
- **Small sample size ($n < 30$)**
- Data is approximately **normally distributed**
- Testing **means of populations**

Type of t-test

One-sample t-test

Independent two-sample t-test

Paired t-test

Purpose

Compare sample mean with population mean

Compare means of two independent groups

Compare means of the same group before and after treatment

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = population mean
- s = sample standard deviation
- n = sample size

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$$

Decision Rule

If $|t \text{ calculated}| > t \text{ critical} \rightarrow \text{Reject } H_0$

If $|t \text{ calculated}| \leq t \text{ critical} \rightarrow \text{Fail to reject } H_0$

t critical values

Degrees of Freedom (df)	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1	6.314	12.706	63.657
2	2.920	4.303	9.925
3	2.353	3.182	5.841
4	2.132	2.776	4.604
5	2.015	2.571	4.032
6	1.943	2.447	3.707
7	1.895	2.365	3.499
8	1.860	2.306	3.355
9	1.833	2.262	3.250
10	1.812	2.228	3.169
12	1.782	2.179	3.055
15	1.753	2.131	2.947
20	1.725	2.086	2.845
25	1.708	2.060	2.787
30	1.697	2.042	2.750
∞	1.645	1.960	2.576

Degrees of freedom df determine which **row of the t-distribution table** to use.

$df = n - 1$ for a one-sample t-test.

$df = n_1 + n_2 - 2$ for two independent samples

As **degrees of freedom increase**, the t-distribution approaches the **Standard Normal Distribution**.

The last row (∞) corresponds to **Z-critical value**

Z test vs t test		
Feature	Z-Test	t-Test
Population variance	Known	Unknown
Sample size	Large (≥ 30)	Small (< 30)
Distribution used	Normal distribution	t-distribution
Variability	Fixed	Depends on sample

The t test:

Hypothesis Test	t-Test Formula	Parameters
Population Mean (One-sample t-test)	$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$	$\bar{x}$ =sample mean, μ = population mean, s = sample standard deviation, n = sample size
Difference Between Two Means (Dependent data)	$t = \frac{\bar{d} \sqrt{n}}{s}$	$\bar{d}$ = mean of difference of sample means, s = standard deviation of difference of sample means
Difference Between Two Means (Independent data)	$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $t = \frac{\bar{x}_1 - \bar{x}_2}{s} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$	$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$ $s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1}$ $s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1}$

Example (Population Mean): The prices of shares in ₹ of a company on different days of a month was found as : 66, 65, 69, 70, 69, 71, 70, 63, 64, 68. Test whether the mean price of shares in the month is ₹65.

$$H_0: \mu = 65$$

The true mean price of the company's shares is ₹65.

$$H_1: \mu \neq 65$$

The true mean price of the company's shares is **different from ₹65.**

$$\bar{x} = A + \frac{\sum d}{n} = 65 + \frac{25}{10} = 67.5$$

$$s = \sqrt{\frac{\sum d^2}{n-1} - \frac{(\sum d)^2}{n(n-1)}} = \sqrt{\frac{133}{9} - \frac{(25)^2}{10 \times 9}} = 2.8$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{67.5 - 65}{2.5/\sqrt{10}} = 2.81$$

$$df = n - 1 = 9,$$

$$t_{critical} = 2.262 (\alpha = 0.05)$$

| t calculated | > t critical → Reject H₀

Hence, average price of shares is not ₹65

SL No	x	$d = x - 65$	d^2
1	66	1	1
2	65	0	0
3	69	4	16
4	70	5	25
5	69	4	16
6	71	6	36
7	70	5	25
8	63	-2	4
9	64	-1	1
10	68	3	9
Σ		25	133

Example (difference between two means for independent data): A company is interested in knowing if there is a significant difference in the average salary received by foremen in two divisions. Accordingly samples of 12 foremen in first division and 10 foreman in second division are selected at random. Based upon experience foremen's salaries are known to be approximately normally distributed and standard deviation are about the same.

	First Division	Second Division
Average weekly salary	10500	9800
Standard Deviation	680	740

Hypothesis:

$H_0: \mu_1 = \mu_2$ There is no significant difference between the average salaries of foremen in the two divisions.

$H_1: \mu_1 \neq \mu_2$ There is a significant difference between the average salaries of foremen in the two divisions.

$$s = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(12-1)680^2 + (10-1)740^2}{12+10-2}} = 707.63$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s} \sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \frac{10500 - 9800}{707.63} \sqrt{\frac{12 \times 10}{12 + 10}} = 2.315$$

$$df = n_1 + n_2 - 2 = 12 + 10 - 2 = 20,$$

$$t_{critical} = 2.086 (\alpha = 0.05)$$

| t calculated | > t critical → Reject H_0

Hence, significant difference in salaries of foreman in two divisions

Example (difference between two means for dependent data): The sales data of an item in six shops before and after a special campaign are in following table. Can the campaign be judged as a success. Test at 5% level of significance.

Hypothesis:

$$H_0: \mu_1 = \mu_2$$

There is **no significant difference** between the mean sales **before and after** the campaign.

The campaign has **no significant effect**.

$$H_1: \mu_1 \neq \mu_2$$

There is a **significant difference** between the mean sales before and after the campaign.

The campaign has a **significant effect**.

Shops	A	B	C	D	E	F
Before Campaign	53	28	31	48	50	42
After Campaign	58	29	30	55	56	45

$$\bar{d} = \frac{\sum d}{n} = \frac{21}{6} = 3.5$$

$$s = \sqrt{\frac{\sum d^2}{n-1} - \frac{(\sum d)^2}{n(n-1)}} = \sqrt{\frac{121}{5} - \frac{(21)^2}{6 \times 5}} = 3.082$$

$$t = \frac{\bar{d} \sqrt{n}}{s} = \frac{3.5 \sqrt{6}}{3.082} = 2.78$$

$$df = n - 1 = 5$$

$$t_{critical} = 2.571 (\alpha = 0.05)$$

| t calculated | > t critical → Reject H₀

Hence, campaign is a success

Shop	X ₁	X ₂	d	d ²
A	53	58	5	25
B	28	29	1	1
C	31	30	-1	1
D	48	55	7	49
E	50	56	6	36
F	42	45	3	9
n=6			∑d=21	∑d ² = 121

CHI SQUARE TEST

The chi square (χ^2) test is a statistical hypothesis test used to determine whether there is a **significant difference between observed frequencies and expected frequencies** in categorical data.

It is widely used for **testing relationships between categorical variables**.

Conditions to apply chi square (χ^2) test

- Data is **categorical**
- Sample observations are **independent and random**
- Expected frequency in each cell is ≥ 5

The **Chi-square test checks whether the difference between observed and expected frequencies is due to chance or indicates a real relationship between variables**.

The **frequency distribution** of variables are arranged in rows and columns called contingency table

A **contingency table** is a **tabular representation of observed frequencies of categorical data** used in the **Chi-Square Test** to study the relationship between two variables.

The rows and columns belong to two categories, say A and B, under study.

Contingency Table (r X c)				
Category	Column 1	Column 2	...	Total
Row 1	O_{11}	O_{12}	...	Row Total R_1
Row 2	O_{21}	O_{22}	...	Row Total R_2
...	...	...	...	...
Total	Column Total C_1	Column Total C_2	...	Grand Total N

O_{ij} = Observed frequency in row i , column j

The expected frequencies are calculated for each cell

$$E_{ij} = \frac{R_i C_j}{N}, \text{ N is total number of observations}$$

chi square (χ^2) is calculated as,

$$\chi^2 = \sum \frac{(O-E)^2}{E} \text{ where O is observed frequency and E is expected frequency}$$

Degree of freedom is calculated as

$$df = (r - 1)(c - 1)$$

Null hypothesis H_0 is that two categories are independent.

Decision rule

If $\chi^2 \text{ calculated} > \chi^2 \text{ critical} \rightarrow \text{Reject } H_0$

If $\chi^2 \text{ calculated} \leq \chi^2 \text{ critical} \rightarrow \text{Accept } H_0$

Example: A sample of 200 persons with a particular disease was selected. Out of them 100 were given a drug and the other were not given any drug. The results are as follows. Test whether the drug is effective or not.

Number of Persons			
	Drug	No Drug	Total
Cured	65	55	120
Not cured	35	45	80
Total	100	100	200

Hypothesis H_0 : two categories are independent i.e. drug is not effective to cure disease

Calculate expected values $E_{ij} = \frac{R_i C_j}{N}$

$$E_{11} = \frac{R_1 C_1}{N} = \frac{120 \times 100}{200} = 60$$

$$E_{12} = \frac{R_1 C_2}{N} = \frac{120 \times 100}{200} = 60$$

$$E_{21} = \frac{R_2 C_1}{N} = \frac{80 \times 100}{200} = 40$$

$$E_{22} = \frac{R_2 C_2}{N} = \frac{80 \times 100}{200} = 40$$

Number of Persons (Expected Frequencies)			
	Drug	No Drug	Total
Cured	60	60	120
Not cured	40	40	80
Total	100	100	200

O	E	$(O - E)^2$	$(O - E)^2 / E$
65	60	25	0.417
35	40	25	0.625
55	60	25	0.417
45	40	25	0.625

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 2.084$$

$$df = (2 - 1)(2 - 1) = 1$$

$$\chi^2 \text{ critical} = 3.84 \text{ (} df = 1, \alpha = 0.05 \text{)}$$

$$\chi^2 \text{ calculated} < \chi^2 \text{ critical} \rightarrow \textbf{Accept } H_0$$

Hence, drug is not effective

ANOVA/F TEST

ANOVA, stands for Analysis of Variance.

It is a statistical method that determines whether significant differences between the averages of three or more unrelated groups.

In addition, this technique is especially useful when comparing more than two groups, which is a limitation of other tests like the t-test and z-test

Use ANOVA when:

- There are **3 or more groups**
- Data is **numerical**
- Groups are **independent**
- Data is approximately **normally distributed**

TYPES OF ANOVA

One-Way ANOVA

- One-way indicates one independent variable
- The One-Way ANOVA analyzes the impact of one single factor on a particular outcome
- Example: Comparing marks of students from 3 different schools

Two-Way ANOVA

- Two- way indicates two independent variables
- Two-Way ANOVA, also known as factorial ANOVA, allows to examine the effect of two different factors on an outcome simultaneously
- Example: Effect of **teaching method + gender** on marks

Hypotheses

- **Null hypothesis (H_0):** All group means are equal
- **Alternative hypothesis (H_1):** At least one mean is different

ANOVA checks:

$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}}$$

Two degree of freedom

- **df_1 (Numerator degrees of freedom) or between-groups df**

$$df_1 = k - 1$$

- Where: k = number of groups
- It represents variation **between the group means**
- **df_2 (Denominator degrees of freedom) or within-groups df**

$$df_2 = N - k$$

- Where, N = total number of observations, k = number of groups
- It represents variation **within the groups**

The **p-value (probability value)** tells you how likely it is to get your observed result **if the null hypothesis (H_0) is actually true.**

Small p-value (≤ 0.05)

Strong evidence **against H_0**
Reject the null hypothesis

Large p-value (> 0.05)

Weak evidence against H_0
Fail to reject H_0 i.e, accept H_0

Decision rule

If **$F > \text{table value at } 0.05 \rightarrow p < 0.05$**

If **$F > \text{table value at } 0.01 \rightarrow p < 0.01$ (very significant)**

H_0 : All group means are equal

If **$p \leq 0.05 \rightarrow$ At least one group mean is different**

If **$p > 0.05 \rightarrow$ No significant difference**

Thank
you